

Speed Detection Camera System (SDCS)

Computer Vision Project Report

Real-Time Vehicle Speed Detection and Violation Monitoring

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1 Abstract

This report presents a comprehensive Speed Detection Camera System (SDCS) designed to automatically monitor vehicle speeds and detect traffic violations in real-time. The system employs a modular computer vision pipeline consisting of four core components: motion detection, foreground segmentation, multi-object tracking, and speed computation. The implementation, based on the methodology proposed by Ibrahim et al. [1], utilizes a hybrid approach combining adaptive background modeling with three-frame differencing to handle sudden illumination changes and stationary objects. The system was evaluated across diverse scenarios including day/night conditions and highway traffic environments. A user-friendly web interface built with Streamlit enables easy video upload, real-time processing visualization, and comprehensive violation reporting.

2 Introduction

2.1 Problem Statement

Traffic speed monitoring is a critical component of road safety enforcement. Traditional manual methods are labor-intensive, prone to human error, and cannot provide 24/7 coverage. Automated speed detection systems using computer vision offer a scalable, consistent, and cost-effective solution for traffic monitoring.

2.2 Objectives

The primary objectives of this project are:

- Develop a fully automated speed detection system using computer vision techniques.
- Implement robust vehicle detection and tracking based on reference research [1].
- Achieve accurate speed measurement through proper camera calibration.
- Detect and document speed violations with photographic evidence.
- Create a user-friendly interface for system operation and result visualization.

2.3 Scope

The system is designed to process pre-recorded video footage from stationary traffic cameras. It focuses on single-lane traffic monitoring with perpendicular camera placement for optimal accuracy. The system handles various environmental conditions including different lighting (day/night) and camera perspectives.

3 Literature Review and Background

3.1 Related Work

Vehicle speed detection has been extensively studied in computer vision literature. Our implementation draws heavily from the "Speed Detection Camera System using Image Processing Techniques" [1], which proposes a hybrid algorithm combining adaptive background subtraction with three-frame differencing.

Common approaches include:

- **Background Subtraction Methods:** Gaussian Mixture Models (GMM), adaptive background learning.
- **Deep Learning Approaches:** YOLO, Faster R-CNN for vehicle detection.
- **Tracking Algorithms:** Kalman filtering, Hungarian algorithm for data association.

4 System Architecture

4.1 Overall Pipeline

The SDCS follows a modular four-stage pipeline:

1. **Motion Detection:** Identifies moving objects using a hybrid of adaptive background modeling and three-frame differencing.
2. **Segmentation:** Separates individual vehicles from the foreground mask.
3. **Tracking:** Associates detections across frames and maintains trajectories.
4. **Speed Capture:** Computes speeds and captures violation evidence.

5 Implemented Algorithms

5.1 Motion Detection Module

5.1.1 Three-Frame Differencing

To improve detection accuracy and reduce false positives from stationary objects, a three-frame differencing algorithm was implemented as described in [1]. This technique compares the current frame with both the previous and next-to-last frames.

Algorithm 1 Three-Frame Differencing Algorithm**Require:** Current Frame I_n , Previous Frame I_{n-1} , Next-to-last Frame I_{n-2} **Ensure:** Motion Mask M_{motion}

```

1:  $D_1 \leftarrow |I_n - I_{n-1}|$  ▷ Difference with previous frame
2:  $D_2 \leftarrow |I_n - I_{n-2}|$  ▷ Difference with next-to-last frame
3: for each pixel  $(x, y)$  do
4:   if  $D_1(x, y) > T_{motion}$  and  $D_2(x, y) > T_{motion}$  then
5:      $M_{motion}(x, y) \leftarrow 1$  ▷ Pixel is moving
6:   else
7:      $M_{motion}(x, y) \leftarrow 0$  ▷ Pixel is stationary
8:   end if
9: end for
10: return  $M_{motion}$ 

```

5.1.2 Adaptive Background Modeling

The final foreground mask is obtained by combining the motion mask from three-frame differencing with adaptive background subtraction to handle shadows and extract the entire object shape.

Algorithm 2 Adaptive Background Detection with Shadow Removal**Require:** Frame I_t , Background B_{t-1} , Threshold T_{t-1} **Ensure:** Foreground mask M_t

```

1: Initialize:  $B_0 \leftarrow I_0$ ,  $T_0 \leftarrow \text{initial\_threshold}$ 
2:  $D_t \leftarrow |I_t - B_{t-1}|$  ▷ Absolute difference
3:  $M_{temp} \leftarrow D_t > T_{t-1}$  ▷ Initial foreground
4: for each pixel  $(x, y)$  do
5:    $ratio \leftarrow I_t(x, y) / B_{t-1}(x, y)$ 
6:   if  $\text{shadow\_min} < ratio < \text{shadow\_max}$  then
7:      $M_t(x, y) \leftarrow 0$  ▷ Mark as shadow
8:   end if
9: end for
10:  $M_t \leftarrow \text{morphological\_operations}(M_t)$ 
11:  $B_t \leftarrow \alpha \cdot I_t + (1 - \alpha) \cdot B_{t-1}$  ▷ Update background
12: return  $M_t$ 

```

5.2 Segmentation Module**5.2.1 Iterative Region Splitting**

The segmentation module uses an iterative horizontal and vertical scanning approach to handle fragmented objects:

Algorithm 3 Iterative Region Splitting Segmentation**Require:** Foreground mask M **Ensure:** List of vehicle regions R

```

1:  $R \leftarrow \text{connected\_components}(M)$ 
2: for  $i = 1$  to  $\text{max\_iterations}$  do
3:    $R_{\text{new}} \leftarrow \emptyset$ 
4:   for each region  $r \in R$  do
5:      $H \leftarrow \text{horizontal\_scan}(r)$  ▷ Split horizontally
6:     for each sub-region  $h \in H$  do
7:        $V \leftarrow \text{vertical\_scan}(h)$  ▷ Split vertically
8:        $R_{\text{new}} \leftarrow R_{\text{new}} \cup V$ 
9:     end for
10:  end for
11:  if  $R_{\text{new}} = R$  then
12:    break ▷ No more splits, converged
13:  end if
14:   $R \leftarrow R_{\text{new}}$ 
15: end for
16: return  $R$ 

```

5.3 Tracking Module

5.3.1 Kalman Filter-Based Tracking

The tracking module uses Kalman filtering for state estimation and trajectory smoothing. The state vector $x = [c_x, c_y, v_x, v_y]^T$ tracks position and velocity.

6 Implementation Details

6.1 Technology Stack

Table 1: Technology Stack and Libraries

Component	Technology/Library
Programming Language	Python 3.11
Computer Vision	OpenCV 4.9.0
Numerical Computing	NumPy 1.26.0
Web Interface	Streamlit 1.32.0
Configuration	PyYAML 6.0

7 Experimental Results and Analysis

7.1 Overall Performance

Table 2: Overall System Performance Metrics

Metric	Our System	YOLOv8	Simple Tracking
Precision	0.89	0.94	0.89
Recall	0.86	0.91	0.84
F1-Score	0.875	0.925	0.865
Speed MAE (km/h)	3.2	2.8	3.9
Processing FPS	18.5	12.3	22.4

8 Work Division

The project responsibilities were distributed among the team members to ensure modular development and efficient integration.

Table 3: Team Member Contributions

Member	Responsibilities	Contribution
Ziad Montaser	Motion Detection Module: Background modeling, three-frame differencing, shadow removal, morphological operations	25%
Abdallah Safan	Segmentation Module: Connected components, iterative splitting, region filtering, merging logic	25%
Ahmed Kamal	Tracking Module: Kalman filter implementation, data association, trajectory management, smoothing	25%
Yara Farouk	Speed Capture Module: Speed calculation, violation detection, image capture, metadata generation, UI development	25%

8.1 Individual Contributions Detail

Ziad Montaser - Motion Detection:

- Implemented adaptive background subtraction algorithm.
- **Implemented three-frame differencing technique** for robust motion detection.
- Developed shadow detection and removal logic.
- Optimized morphological operations for noise reduction.

- Conducted parameter sensitivity analysis.

Abdallah Safan - Segmentation:

- Designed iterative region splitting algorithm.
- Implemented horizontal and vertical scanning logic.
- Developed region merging for fragmented detections.
- Created segmentation visualization tools.
- Optimized segmentation parameters for various scenarios.

Ahmed Kamal - Tracking:

- Implemented Kalman filter for state estimation.
- Developed Hungarian algorithm for data association.
- Created trajectory smoothing and validation logic.
- Handled track initialization and termination.
- Implemented ID management and track history.

Yara Farouk - Speed Capture & Integration:

- Implemented speed calculation algorithms.
- Developed violation detection and capture system.
- Created multi-frame image enhancement.
- Built Streamlit web interface.
- Integrated all modules into complete pipeline.

9 Conclusion

This project successfully developed a complete Speed Detection Camera System using classical computer vision techniques, implementing the methodologies discussed in [1]. The system demonstrates that algorithmic approaches, particularly the combination of three-frame differencing and adaptive background subtraction, can achieve competitive performance (87.5% F1-score) while offering significant advantages in processing speed (50% faster than deep learning baselines).

10 References

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